

PRIMARY ISSUE CATEGORY: MUD LOSS

1. PROBLEM DEFINITION

Mud loss (lost circulation) occurs when drilling fluid pumped into the wellbore does not fully return to surface, escaping into the formation through natural fractures, vugs, high-permeability zones, or induced fractures caused by exceeding the formation's fracture/loss gradient.

2. EARLY WARNING SIGNS

- Pit volume decreasing faster than expected while pumps run at a constant rate.
- Flow-out to flow-in ratio dropping below ~0.9.
- Sudden drop in standpipe pressure without a change in pump rate.
- Drilling into a known high-permeability sand streak or naturally fractured/vuggy formation.

3. ROOT CAUSES (FIELD_ALPHA HISTORY)

- Unsupported, high-permeability sand streaks (e.g. Formation_X, 2880-2950 m) with low fracture gradient.
- ECD exceeding the formation's loss threshold, especially during connections or after a mud-weight increase.
- Naturally fractured or vuggy carbonate/shale intervals.
- Depleted or low-pressure zones from prior production in mature fields.

4. STEP-BY-STEP SOLUTION / RESPONSE PROCEDURE

Step 1: Stop and assess — Halt drilling, pull off bottom if safe, and confirm the loss rate by monitoring pit volume over a fixed interval (e.g. 15 min).

Step 2: Reduce ECD — Slow the pump rate and, if losses persist, reduce mud weight in small increments (0.02-0.05 SG) while monitoring well control margins.

Step 3: Spot an LCM pill — Mix and spot a fine-to-medium LCM pill (typically 60 lb/bbl CaCO₃/mica blend, 25-40 bbl) across the loss interval using a balanced-plug technique.

Step 4: Shut in and let the pill set — Allow 1-4 hours of soak time depending on pill design before attempting to circulate through the treated interval.

Step 5: Confirm returns and resume — Circulate bottoms-up at reduced rate to confirm full returns before resuming drilling; escalate to cement plug or casing-off if losses persist after 2 pill attempts.

5. MATERIALS / PARAMETERS QUICK REFERENCE

Loss Severity	Typical LCM Pill	Volume	MW Adjustment
Seepage (<10 bbl/hr)	Fine CaCO ₃ , 40-60 lb/bbl	15-20 bbl	None or -0.02 SG
Partial (10-50 bbl/hr)	Fine/medium CaCO ₃ +mica, 60 lb/bbl	25-40 bbl	-0.03 to -0.05 SG
Severe (>50 bbl/hr)	Coarse LCM / cross-linked pill	40-60 bbl	-0.05 SG or more, consider casing off

6. PREVENTION CHECKLIST FOR OFFSET WELLS

- Pre-mix LCM pills before entering any known loss-prone interval (do not wait for losses to start).
- Trim mud weight through confirmed high-permeability streaks where well control margin allows.
- Increase flow-out monitoring frequency (every 5 min) through loss-risk depth windows.
- Flag loss-prone intervals in the NWIS early-warning model based on offset-well depth correlation.

7. FIELD CASE REFERENCES

Well	Depth	Outcome
W001	2905 m	Controlled with 35 bbl LCM pill + 0.04 SG MW reduction; 6.5 hr NPT
W002	2885 m	Controlled with 25 bbl LCM pill + 0.04 SG MW reduction; ~2 hr NPT
W003	2920 m	Controlled with 25 bbl LCM pill + 0.04 SG MW reduction; ~1.5 hr NPT

This Issue Resolution Playbook is a synthetically generated reference document produced for the NWIS prototype, consolidating lessons across Field_Alpha and Field_Alpha-adjacent wells. Not associated with an actual drilling operation.